

# CSCI 112

## Programming with C

### LECTURE: 5

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# OUTLINE:

Control flow:

- if-else statement
- switch statement
- goto statement

# CONTROL

## Flow

# IF-ELSE STATEMENT 1/4

```
if (expression1)
        single_statement1
else
        single_statement2
```

- `expression1` is non-zero ⇒ statement1
- `expression1` could be series of expressions
- Block can be substitute for `simple_statement`
- `else` is optional

# IF-ELSE STATEMENT 2/4

```c
int main()
{
    int x = 5;
    if( x > 1 )
        printf("%s\n", "a variable x is greater than 1");
}
```



```
a variable x is greater than 1
```

- Why is an if-else statement considered a statement when it doesn't change anything on its own?
- It's not true that it doesn't change the PC, but to explain that we need to understand ternary operator and labels.

# IF-ELSE STATEMENT 3/4

## With optional `else`

```c
int main()
{
    int x = 5;
    if( x > 1 )
        printf("%s\n", "a variable x is greater than 1");
    else
        printf("%s\n", "a variable x is NOT greater than 1");
}
```

```
a variable x is greater than 1
```

# IF-ELSE STATEMENT 4/4

```c
int main()
{
    int x = 0;
    if( x > 1 )
    {
      printf("%s", "a variable x");
      printf("%s\n"," is greater than 1");
    }
    else
    {
      printf("%s", "a variable x");
      printf("%s\n"," is not greater than 1");
    }
}
```



```
a variable x is not greater than 1
```

# NESTED `IF-ELSE` STATEMENT 1/3

- Since an `if` statement following its condition in parentheses (the expression that, if it evaluates to a non-zero value, executes the subsequent code block) is considered a statement itself;
- it's important to note that another if statement can be placed after either the code block of the first `if` or its `else` block (if present).

# NESTED IF-ELSE STATEMENT 2/3

```c
int main()
{
  int x = 1;

  if(x == 0)
  ;
  else if(x == 1)
  ;
  else if(x > 2)
  ;
  else
  ;
}
```

```c
int main()
{
  int x = 1;
  if(x == 0)
  {
  }
  else
  if(x == 1)
  {
  }
  else
  if(x > 2)
  {
  }
  else
  {
  }
}
```

# NESTED IF-ELSE STATEMENT 3/3

```c
int main()
{
   int x = getche() - 48;
   printf("%s","a variable x is ");

   if(x == 0)
   {
     printf("%s\n","0");
   }
   else
   {
     if(x == 1)
     {
          printf("%s\n","1");
     }
     else
     {
       if(x > 1 && x < 5)
       {
          printf("%s\n","between 2 and 4");
       }
       else
       {
         printf("%s\n","not between 0 and 4");
       }
     }
   }
}
```

```c
int main()
{
  int x = getche() - 48;
  printf("%s","a variable x is ");

    if(x > 0)
    if(x < 5)
      printf("%s\n", "between 1 and 4");

}
```

# SWITCH

## statement

# SWITCH STATEMENT 1/1

```
switch (expression)
{
 case <constant-expression1>:
 /* code */
  break;

 case <constant-expression2>:
 /* code */
  break;

 default:
 /* code */
  break;
}
```



```
int main()
{
  int x = getche() - 48 ; // typed 4
  printf("%s","a variable x is ");
  switch (x)   /* constant-expression */
  {
    case 1:
      printf("%s\n","1");
      break;
    case 2:
      printf("%s\n","2");
      break;
    default:
      printf("%s\n","not between 1 and 3"); // <- true
      break;
  }
} // result: "a variable x is not between 1 and 3"
```

# SWITCH STATEMENT 1/2

In C programming, a **constant expression** is an expression that evaluates to a fixed value at compile time, meaning the value is determined during the compilation process rather than during program execution.

- The `default` block in the switch statement is optional!
- The `default` block can be used for error handling.
- It's important to note that omitting the `default` block will cause the program to skip the entire switch statement if the variable takes on a value not defined in the switch statement.
- To terminate a switch statement, we use the break keyword

# SWITCH STATEMENT 1/3

```c
int main()
{
  int x = getche() - 48; //typed 2
  printf("%s", "a variable x is ");
  switch (x)   /* constant-expression */
  {
    case 1:
      printf("%s\n","1");
      break;
    case 2:
      printf("%s\n","2");
      break;
  }
} // result: "a variable x is 2"
```



# SWITCH STATEMENT 1/4

```c
int main()
{
  int x = getche() - 48;  // typed 2
  printf("%s", "a variable x is ");
  switch (x)   /* constant-expression */
  {
    case 1:
    case 2:
    case 3:
      printf("%s\n", "between 1 and 3");
      break;
    default:
      printf("%s\n", "not between 1 and 3");
      break;
  }
} // result: "a variable x is  1 and 3"
```

```c
int main()
{
  int x = getche() - 48;  // typed 2
  printf("%s", "a variable x is ");
  switch (x)   /* constant-expression */
  {
    case 1:
      printf("%s", "between");
    case 2:
      printf("%s", " 1 and");
    case 3:
      printf("%s\n", " 3");
      break;
    default:
      printf("%s\n", "not between 1 and 3");
      break;
  }
} // result: "a variable x is  1 and 3"
```

# DIFFERENCE BETWEEN `SWITCH` & `IF-ELSE`

- **`if` statements**:
  - **Dynamic Conditions**: if statements allow for dynamic conditions that can change during program execution. These conditions can involve complex expressions, function calls, and even user input.
- **`switch` statements**:
  - **Static Values**: switch statements are primarily designed for evaluating expressions against a set of predefined, static values. These values are typically determined before program execution.

# GOTO

## statement

# GOTO STATEMENT 1/2



```c
int main()
{
  bool is_true = true;
  if ( is_true )
    goto label1;
  else
    goto label2;

  label1:
    printf("It was true\n");
    goto end;
  label2:
    printf("It was false\n");
    goto end;
  end:
} // result: It was true
```

# GOTO STATEMENT 2/2

- goto allows us to jump (within the same function) unconditionally to a location identified by a `label`
- a simple example of Spaghetti Code - Dijkstra, E. W. (1968)

# TERNARY

## conditional operator

# TERNARY CONDITIONAL OPERATOR ?:

- If `expression1` evaluates to `true`, the ternary operator returns the value of `expression2`. Otherwise, it returns the value of `expression3`.

```
<expression1> ? <expression2>: <expression3>;
```

```c
int main()
{
  int i = 1;
  i = (i > 0) ? 2 : 3;        /* ternary conditional */
  printf("%d\n", i);
}
```

Result:

```
2
```

# IF-ELSE STATEMENT



```
/* 01 */    int main()              /* Program Counter =01;          PC++; “goto” new PC  */
/* 02 */    {                       /* Program Counter =02;          PC++; “goto” new PC  */
/* 03 */        if ( x > 0 )        /* if x > 0 then PC = 04 otherwise PC = 10; “goto” new PC      */
/* 04 */        {                   /* Program Counter =04;          PC++; “goto” new PC  */
/* 05 */            ;               /* Program Counter =05;          PC++; “goto” new PC  */
/* 06 */            ;               /* Program Counter =06;          PC++; “goto” new PC  */
/* 07 */            ;               /* Program Counter =07;          PC++; “goto” new PC  */
/* 08 */        }                   /* Program Counter =08;          “goto” PC=15         */
/* 09 */        else                /*                                    */
/* 00 */        {                   /* Program Counter =10;          PC++; “goto” new PC  */
/* 11 */            ;               /* Program Counter =11;          PC++; “goto” new PC  */
/* 12 */            ;               /* Program Counter =12;          PC++; “goto” new PC  */
/* 13 */            ;               /* Program Counter =13;          PC++; “goto” new PC  */
/* 14 */        }                   /* Program Counter =14;          PC++; “goto” new PC  */
/* 15 */    }                       /* Program Counter =15;          PC++; “goto” new PC  */
```

# MAGIC...

```c
int main()
{
/* 01 */            int x = 10;
/* 02 */            int * label_pointer;
/* 03 */    label_pointer = ( x > 0 ) ? &&label_greater: &&label_lower; /* equals 08 or 12 */
/* 04 */            goto *label_pointer;
/* 05 */    label_greater:
/* 06 */            printf("%s\n","a variable x is greater than 0");
/* 07 */            goto end_label;
/* 08*/     label_lower:
/* 09 */            printf("%s\n","a variable x is lower than 0");
/* 11 */            goto end_label;
/* 12 */    end_label:
/* 13 */            ;
}
```



# THANK

You